



Survey on Factuality in Large Language Models

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This survey addresses the crucial issue of factuality in Large Language Models (LLMs). As LLMs find applications across diverse domains, the reliability and accuracy of their outputs become vital. We define the “factuality issue” as the probability of LLMs to produce content inconsistent with established facts. We first delve into the implications of these inaccuracies. Subsequently, we analyze the mechanisms through which

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LLMs store and process facts, seeking the primary causes of factual errors. Our discussion then transitions to methodologies for evaluating LLM factuality, emphasizing key metrics, benchmarks, and studies. We further explore strategies for enhancing LLM factuality. Our survey offers a structured guide for researchers aiming to fortify the factual reliability of LLMs. We consistently maintain and update the related open-source materials at <https://github.com/wangcunxiang/LLM-Factuality-Survey>.

CCS Concepts: • **Computing methodologies** → **Natural language generation**;

Additional Key Words and Phrases: Factuality, large language models, knowledge, retrieval augmentation

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1 Introduction

The pursuit of knowledge mastery has been a fundamental aspiration driving the development of artificial intelligence systems. Historically, seminal works by McCarthy et al. [118] and Newell and Simon [131] have underscored the significance of knowledge representation and reasoning in AI systems. For instance, the Cyc project embarked on an ambitious journey to codify common-sense knowledge, aiming to provide AI systems with a comprehensive understanding of the world [89]. Concurrently, endeavors like the WordNet project by Miller et al. [124] sought to create lexical databases that capture semantic relationships between words, thereby aiding AI systems in grasping the nuances of natural language.

Amidst these pioneering efforts, the emergence of **Large Language Models (LLMs)**, such as ChatGPT [133], GPT-4 [134], and LLaMA [174, 175], has been seen as a significant leap in both academics and industries, particularly in developing AI systems with vast factual knowledge [31, 134, 142]. The advantages of using LLMs as knowledge bases carriers are manifold. Firstly, they reduce the overhead and costs associated with building and maintaining dedicated knowledge bases [2, 144, 184]. Additionally, LLMs offer a more flexible approach to knowledge processing and utilization, allowing for context-aware reasoning and the ability to adapt to novel information or prompts [64, 166]. Yet, with their unparalleled capabilities, concerns have arisen about the potential of LLMs to generate non-factual or misleading content [7, 10, 134]. In light of these advancements and challenges, this survey seeks to delve deeply into the LLMs, exploring both their potential and the concerns surrounding their factual accuracy.

Understanding the factuality of LLMs is more than just a technical challenge; it's essential for the responsible use of these tools in our daily lives. As LLMs become more integrated into services like search engines [122], chatbots [55, 133], and content generators [27], the information they provide directly influences decisions, beliefs, and actions of millions of people. If an LLM provides incorrect or misleading information, it can lead to misunderstandings, spread false beliefs, or even cause harm, especially for those domains that demand high factual accuracy [99], such as health [172, 173], law [65], and finance [193]. For instance, a physician relying on an LLM for medical guidance might inadvertently jeopardize patient health, a corporation leveraging LLM insights might make ill-informed market decisions, or an attorney misinformed by an LLM might falter in legal proceedings [28]. In addition, with the advancement of LLM-based agents, the factuality of LLMs is becoming even more potent. A driver or an autonomous driving car might rely on LLM-based agents for planning or driving, where serious factual mistakes made by LLMs could cause

irreversible damage. By studying the factuality of LLMs, we aim to ensure that these models are not only powerful but also reliable and trustworthy.

A surge of research has been directed towards evaluating LLMs' factuality, which encompasses diverse tasks like factoid question answering and fact checking. Beyond evaluation, efforts to improve the factual knowledge of LLMs have been notable. Strategies have ranged from retrieving information from external knowledge bases to continual pretraining and supervised finetuning. Yet, despite these burgeoning efforts, a holistic overview that covers the full spectrum of factuality in LLMs remains elusive. To the best of our understanding, our work is the first comprehensive study on the factuality of LLMs.

This survey aims to offer an exhaustive overview of the factuality studies in LLMs, delving into four key dimensions: Section 2) The definition and impact of the factuality issue [132, 146]; Section 3) Techniques for evaluating factuality and its quantitative assessment [66, 125]; Section 4) Analyzing the underlying mechanisms of factuality in LLMs and identifying the root causes of factual errors [85, 103]; and Section 5) Approaches to enhance the factuality of LLMs [37, 59]. Notably, we categorize the use of LLMs into two primary settings: LLMs without external knowledge, such as ChatGPT [133] and Retrieval-Augmented LLMs, such as BingChat [122]. The complete structure of this survey is illustrated in Figure 1. Through a detailed examination of existing research, we seek to shed light on this critical aspect of LLMs, helping researchers, developers, and users harness the power of these models responsibly and effectively. An extended version of this article is uploaded to the arxiv in <https://arxiv.org/abs/2310.07521>.

2 Factuality Issue

In this section, we describe the issue of factuality in LLMs.

2.1 LLMs

There is no universally accepted or precise definition of LLMs in the literature [12, 41, 64]. We primarily focus on the pre-trained language models with emergent decoder-only generative modes, such as ChatGPT [133] and LLaMA [174, 175]. We also include some work based on models with encoder-decoder architectures, such as T5 [151]. In this survey, we do not talk about work that is based only on encoder-only models, such as BERT [33] and RoBERTa [107].

2.2 Factuality

Factuality in LLMs refers to their capability of LLMs for generating contents that adheres to factual information, which encompasses commonsense, world knowledge and domain facts. The factual information can be grounded to reliable sources, such as dictionaries, Wikipedia or textbooks from different domains.¹ A series of work have discussed whether LLMs can serve as knowledge bases to store factual knowledge [2, 136, 203]

Existing works focus on measuring factuality in LLMs qualitatively [19, 97], discussing the mechanism for storing knowledge [18, 120] and tracing the source of knowledge issues [56, 80]. The factuality issue for LLMs receive relatively the most attention. Several instances are shown in Table 1. For instance, an LLM might be deficient in domain-specific factual knowledge, such as medicine or law domain. Additionally, the LLM might be unaware of facts that occurred post its last update. There are also instances where the LLM, despite possessing the relevant facts, fails to reason out the correct answer. In some cases, it might even forget or be unable to recall facts it has previously

¹We only consider cases where the meaning is clear and the truthfulness can be determined in this survey. Furthermore, we only take into account undisputed facts. Minor errors that may exist in reliable sources are not within the scope of our consideration.

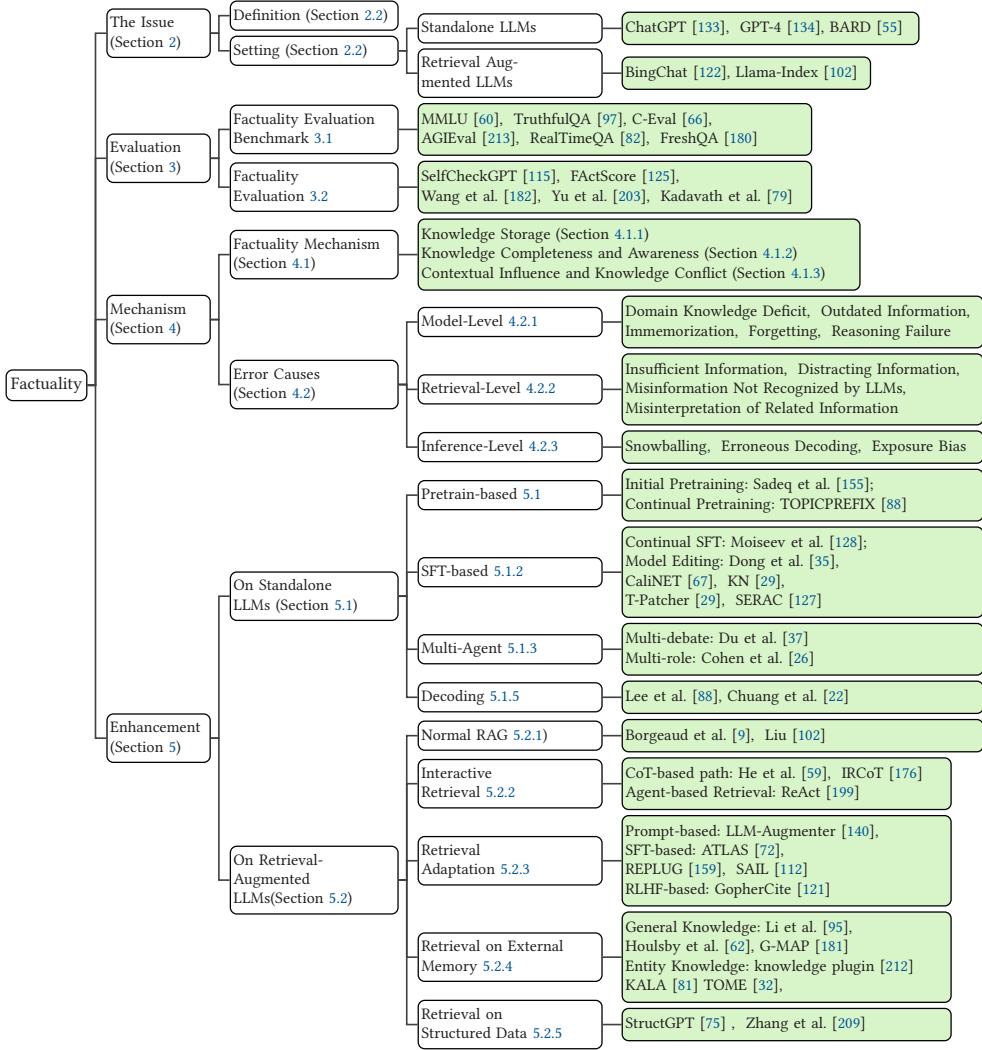


Fig. 1. Taxonomy of research on factuality in LLMs that consists of the issue, evaluation, analysis and enhancement.

learned. The factuality problem is closely related to several hot topics in the field of LLMs, including Hallucinations [41, 73], Outdated Information [129, 149], and Domain-Specificity (e.g., Health [186, 195], Law [27], and Finance [193]). At their core, these topics address the same issue: the potential for LLMs to generate contents that contradicts certain facts, whether those contents arise out of thin air, outdated information, or a lack of domain-specific knowledge. To provide a clearer structure to the comparison of factuality, hallucination, and outdated information in LLMs, we provide explanation for each term followed by how they relate to the broader factuality problem (also shown in Table 2):

- **Factuality:** Factuality refers to the ability of an LLM to provide information that is accurate, verifiable, and aligned with real-world knowledge. Factuality concerns focus on the correctness of content in relation to known facts or databases.

Table 1. Examples of Different Kinds of Factual Error Produced by LLMs

Category	Cause	Example Dialog	Notes and references
Model-level causes	Domain knowledge deficit	Q: CEO of Assicurazioni Generali? BloombergGPT: Philippe Donnet GPT-NeoX: Antonio De Lorenzo, Simone Gambarini, Enrico Zanetti FLAN-T5-XXL: John M Forsyth, Christopher K Peters, [empty string]	BloombergGPT is a finance domain-specific language model. [193]
	Outdated information	Q: When was Kyiv attacked by Russia? ChatGPT: As of my last knowledge update in September 2021, Russia had not launched an attack on Kyiv.	Kyiv was attacked by Russia on 25 February 2022.
	Reasoning error	Q: Who is Tom Cruise's mother? A: Mary Lee Pfeiffer Q: Who is Mary Lee Pfeiffer's son? A: There is no widely known information about...	From Berglund et al. [8]. It is clear that the model knows Tom Cruise's mother is Lee Pfeiffer, but it fails to reason that Lee Pfeiffer has a son named Tom Cruise.
Retrieval-level causes	Distracted by the retrieval information	Retrieved context: "Alan Melikdjanian - Age, Bio, Faces and Birthday" Melikdjanian was the Director, Writer (with his father), Editor, and Composer of the 2006 direct-to-DVD Russian-language film Citizen Mavzik, the story of a Russian immigrant family's assimilation into American culture. The film was produced by Vilalan Productions, named after Melikdjanian and his father Vilen. Q: What is the nationality of the director of Citizen Mavzik? A: (1) The director of the film Citizen Mavzik is Alan Melikdjanian. (2) Alan Melikdjanian is from Russia. So, the answer is Russia.	Alan Melikdjanian, known by the alias Captain Disillusion, is a Soviet-born American independent filmmaker and YouTuber [119].
	Misunderstood the retrieval information	Retrieved context: "Gerard III, Count of Guelders - Wikipedia" Gerard married Margaretha of Brabant, the daughter of Duke Hendrik I of Brabant and Matilda of Flanders, Duchess of Brabant, in 1206. Gerard and Margaret's Tomb ... Q: Who is Gerard I, Count Of Guelders's wife? A: ... the name of Gerard I, Count of Guelders's wife, which is Margaretha of Brabant.	The retrieved context accurately provided the name of Gerard III, Count of Guelders' wife, which is Margaretha of Brabant. However, the model erroneously referred to him as Gerard I, resulting in an incorrect answer.
	Fail to address misinformation in retrieved documents	Retrieved context: [0] "James Vincent - Death Notice and Service Information - Legacy.com" James Vincent died on August 9, 2023 in Long Beach, California Obituary. James Joseph Vincent was the perfect husband, father, grandfather... [1] "UPDATE: Jury convicts James Vincent of murder after deliberating" A jury convicted James Vincent of murder in the 2018 shooting death of Luka Grabb. The jury reached its verdict at 4:30 p.m. Friday. Q: When did James Vincent die? A: Based on the information provided in the two articles, James Vincent died on August 9, 2023, in Long Beach, California.	In the presence of contradictory information from retrieved documents, the model should thoroughly outline the response, while also noting the need for clarity. For instance, the model could state: "According to the first document, James Vincent passed away on August 9, 2023, in Long Beach. However, the second document suggests that James Vincent was murdered in 2018."
	Snowballing	Q: Was there ever a US senator who represented the state of New Hampshire and whose alma mater was the University of Pennsylvania? A: Yes... His name was John P. Hale Q: Aamari was very stoic. [PRONOUN] rarely showed any emotion. A: He.	John P. Hale was graduated from Bowdoin College. [208] The correct answer was Xe according to [61].

We category the factual error types by the causes of them, whose details can be found in Section 4.2.

Table 2. Comparison of Hallucination, Factuality, Outdated Information, and Domain-Specificity

Aspect	Factuality	Hallucination	Outdated Information	Domain-Specificity
Focus	Accuracy in relation to known, verifiable information	Unwarranted or irrelevant content generation	Use of information that has since become incorrect	Content requiring specialized knowledge
Example	Misquoting a scientific fact	Story about a dog instead of a wolf (not from prompt)	Using political facts from 2021 that changed in 2023	Technical errors in a medical research response
Relation to Factuality	Directly impacts the truthfulness of generated content	May produce factually correct but prompt-divergent content	Leads to inaccuracies due to reliance on old data	Factual errors in specialized areas, requiring deep expertise

— **Hallucination:** Hallucination occurs when an LLM generates content that is either nonsensical or untruthful with respect to the input or underlying knowledge sources. Hallucination does not always imply factual inaccuracy but rather divergence from expected content or the input prompt.

- **Outdated Information:** Outdated information arises when an LLM generates responses based on knowledge that was once accurate but has since been superseded by more recent facts.
- **Domain-Specificity:** Domain-specificity involves the model generating content that requires specialized or technical knowledge in particular fields.

Setting. In this survey, our primary focus is on two specific settings:

- (1) Standard LLMs: Directly using LLMs for answering and chatting [133, 134];
- (2) A Retrieval-Augmented LLM: an ordinary LLM with retrieval-augmented generation [102, 122].

The latter is of particular interest as retrieval mechanisms are among the most prevalent methods to enhance the factuality of LLMs. This involves not just generating accurate responses, but also correctly selecting pertinent knowledge snippets from the myriad of retrieved sources.

While summarization tasks—where the goal is to produce summaries that stay true to the source input—have seen research on factuality [117, 170], we opted not to focus heavily on this domain in our survey for a few reasons. Firstly, the source inputs for summarization often contain content that is not factual. Secondly, summarization introduces unique challenges, such as ensuring coherence, conciseness, and relevance, which deviate from the focus of this survey. It is also worth noting that Pu et al. [147] found that LLMs produce fewer factual errors or hallucinations compared to humans in various summarization benchmarks. However, we will still discuss some works in this area, particularly those that overlap with retrieval settings.

3 Factuality Evaluation

Evaluating the factuality of LLMs is pivotal for ensuring the reliability and trustworthiness of their generated content [88, 145]. As LLMs become increasingly integrated into various applications, ranging from information retrieval to content generation, the accuracy of their outputs becomes paramount. In this section, we delve into the evaluation metrics and benchmarks used for assessing the factuality of LLMs and studies that have undertaken such evaluations.

3.1 Benchmarks for Factuality Evaluation

This section examines key benchmarks and metrics used to evaluate the factuality of LLMs. Table 3 presents specific benchmarks designed for assessing LLM factuality, while Table 4 outlines relevant evaluation metrics.

MMLU [60] and TruthfulQA [97] stand as two pivotal benchmarks in the realm of evaluating the factuality of LLMs [134, 139, 175]. The MMLU benchmark is proposed to measure a text model's multitask accuracy across a diverse set of 57 tasks. These tasks span a wide range of subjects, ranging from elementary mathematics to US history, computer science, law, and more. The benchmark is designed to test both the world knowledge and problem-solving ability of models. The findings from the article suggest that while most recent models perform at near random-chance accuracy, the largest GPT-3 model demonstrated a significant improvement. However, even the best models still have a long way to go before achieving expert-level accuracy across all tasks [134]. TruthfulQA is a benchmark designed to assess the truthfulness of a language model's generated answers. The benchmark consists of 817 questions spanning 38 categories, including health, law, finance, and politics. These questions were crafted in such a way that some humans might answer them falsely due to misconceptions or false beliefs. The goal for models is to avoid generating these false answers that they might have learned from imitating human texts.

HaluEval [92] is designed to understand and evaluate the propensity of LLMs like ChatGPT to generate hallucinations. A hallucination, in this context, refers to content that either conflicts with

Table 3. Benchmarks for Evaluating Factuality in LLMs

Reference	Task Type	Dataset	Metrics	Performance of Representative LLMs
MMLU [60]	Multi-Choice QA	Humanities, Social Sciences, STEM...	ACC	(ACC, 5-shot) GPT-4: 86.4 GPT-3.5: 70 LLaMA2-70B: 68.9
C-Eval [66]	Multi-Choice QA	STEM, Social Science, Humanities...	ACC	(zero-shot, average ACC) GPT-4: 68.7 GPT-3.5: 54.4 LLaMA2-70B: 50.13
AGIEval [66]	Multi-Choice QA	Gaokao, (geometry, Bio, history...), SAT, Law...	ACC	(zero-shot, average ACC) GPT-4: 56.4 GPT-3.5: 42.9 LLaMA2-70B: 40.02
PopQA [114]	Multi-Choice QA	PopQA, EntityQuestions	ACC	(overall ACC) GPT-3.5: ~37.0
QUIP [192]	Generative QA	TriviaQA, NQ, ELI5, HotpotQA	QUIP-Score Exact match	(ELI5, QUIP, null prompt) GPT-4: 21.0 GPT-3.5: 27.7
FreshQA [180]	Generative QA	FRESHQA	ACC (Human)	(strict ACC, null prompt) GPT-4: 28.6 GPT-3.5: 26.0
RealTimeQA [82]	Multi-Choice QA, Generative QA	RealTimeQA	ACC, F1	(original setting, GCS retrieval) GPT-3: 69.3 (ACC for MC) GPT-3: 39.4 (F1 for generation)
TruthfulQA [97]	Multi-Choice QA, Generative QA	Health, Law, Conspiracies, Fiction...	Human Score, GPT-judge, ROUGE, BLEU, MC1, MC2...	(zero-shot) GPT-4: ~29 (MC1) GPT-3.5: ~28 (MC1), 79.92(%true) LLaMA2-70B: 53.37 (%true)
HaluEval [92]	Hallucination Evaluation	HaluEval	ACC	(general ACC) GPT-3.5: 86.22
ALCE [49]	Citation Generation	ASQA, ELI5, QAMPARI	MAUVE, Exact Match, ROUGE-L...	(ASQA, 3-psg, citation prec) GPT-3.5: 73.9 LLaMA-33B: 23.0
SelfAware [202]	Self-evaluation	SelfAware	ACC	(instruction input, F1) GPT-4: 75.47 GPT-3.5: 51.43 LLaMA-65B: 46.89
UniLC [211]	Fact Checking	Climate, Health, MGFN	ACC F1	(zero-shot, fact tasks, average F1) GPT-3.5: 51.62
Pinocchio [63]	Fact Checking QA, Reasoning	Pinocchio	ACC F1	GPT-3.5: (Zero-shot ACC: 46.8, F1:44.4) GPT-3.5: (Few-shot ACC: 47.1, F1:45.7)
BigBench [165]	Multi-tasks (QA, NLI, Fact Checking, Reasoning...)	BigBench	Metric to each type of task	(Big-Bench Hard) GPT-3.5: 49.6 LLaMA-65B: 42.6

This table details each benchmark's task type, associated sub-datasets, evaluation metrics, and the performance of select representative LLMs.

the source or cannot be verified based on factual knowledge. The HaluEval benchmark comprises a vast collection of generated and human-annotated hallucinated samples, aiming to evaluate the performance of LLMs in recognizing such hallucinations. The benchmark utilizes a two-step framework, termed “sampling-then-filtering”, based on ChatGPT to generate these samples. Additionally, human labelers were employed to annotate hallucinations in ChatGPT responses.

BigBench [165] focuses on the capabilities and limitations of LLMs. It comprises 204 tasks from diverse domains such as linguistics, childhood development, math, common-sense reasoning, biology, physics, social bias, software development, and more. The benchmark is designed to evaluate tasks believed to be beyond the capabilities of current language models. The study evaluates the

Table 4. Evaluation Metrics for the Factuality of LLMs

Evaluation Type	Metric	Description, Purpose/Usage, and Reference/Findings
Rule-based	Exact Match	Correct matches between generated and input/reference text.
	Common Metrics (Accuracy, Precision, Recall, F-Measure, Calibration, Brier)	Accuracy: Measures overall correctness of model prediction. Precision: Ability of model to classify instances positively. Recall: Ability of model to identify all positive instances. F-Measure: Combines both precision and recall into a single score. Calibration: Agreement between predicted probabilities and observed frequencies. Brier: Measures accuracy of probabilistic predictions.
	MC1 and MC2 [97]	Used to identify the correct answer from multi-choice questions.
	BLEU [138]	Determines frequency of phrase co-occurrence in two sentences.
	ROUGE [96]	Calculates similarity between reference and generated text.
Neural	METEOR[109]	Comprehensive measure based on harmonic mean of unigram precision and recall.
	QUIP-Score [192]	Calculates how much of generated text consists of exact spans found in a text corpus.
	ADEM [109]	Uses a Hierarchical RNN to evaluate the quality of responses in language models.
	BERTScore [210]	Pre-trained embeddings from BERT used to evaluate sentence similarity.
	BLEURT [156]	Trains BERT on larger dataset and rate similarity of sentences.
LLM-based	BARTScore [204]	Evaluates quality using pre-trained sequence-to-sequence models.
	AIS [153], Auto-AIS[48]	Evaluates if output is backed by evidence.
	FActScore [125]	Measures factual accuracy of LLMs breakpointing generated content into atomic facts.
	GPTScore [44]	Evaluates quality of AI output. It is efficient and avoids the annotation requirement.
	GPT-judge [97]	Evaluates truthfulness of LLM answers. It is used in evaluating truthfulness and informativeness.
	Truthfulness and Informativeness [97]	Truthfulness Measures the honesty of LLM information. It is used to evaluate the factuality of information. Informativeness Evaluates the relevance and value of LLM responses
	LLM-Eval [98]	Evaluates the quality of a conversation. It is adaptable in various scenarios.

performance of various models, including OpenAI’s GPT series, on BIG-bench and compares them to human expert raters. Key findings suggest that model performance and calibration improve with scale but still fall short of human-level proficiency.

Huang et al. [66] propose C-Eval, the first comprehensive Chinese evaluation suite. It can be used to evaluate the advanced knowledge and reasoning abilities of foundational models within a Chinese context. The evaluation suite comprises multiple-choice questions spanning 52 diverse disciplines, with four levels of difficulty: middle school, high school, college, and professional. Additionally, C-Eval Hard is introduced for very challenging subjects within the C-Eval suite, which demands advanced reasoning abilities to solve. Their evaluations of state-of-the-art LLMs, including both English and Chinese-oriented models, reveal that there is significant room for improvement as only GPT-4 managed to achieve an average accuracy of over 60%.

SelfAware [202] aims to investigate whether models recognize what they don’t know. This dataset encompasses two types of questions: unanswerable and answerable. The dataset comprises 2,858 unanswerable questions collected from various websites and 2,337 answerable questions extracted from sources such as SQuAD, HotpotQA, and TriviaQA. Each unanswerable question is confirmed as such by three human evaluators.

The Pinocchio benchmark [63] serves as an comprehensive evaluation platform, focusing on assessing factuality and reasoning for LLMs. This benchmark encompasses 20,000 diverse factual queries from diverse sources, timeframes, fields, regions, and languages. It tests an LLM’s capability to discern combined facts, process both structured and unstructured evidence, recognize the temporal evolution of facts, pinpoint minute factual disparities, and withstand adversarial inputs.

Kasai et al. [82] introduce a dynamic QA platform named REALTIMEQA. This platform is unique in that it announces questions and evaluates systems on a regular basis, specifically weekly. The questions posed by REALTIMEQA pertain to current events or novel information, challenging the static nature of traditional open domain QA datasets. The platform aims to address instantaneous information needs, pushing QA systems to provide answers about recent events or developments.

FreshQA [180] is a dynamic benchmark designed to evaluate up-to-date world knowledge of LLMs. Its questions range from those that are never-changing to those that are fast-changing, as well as questions based on false premises. The aim is to challenge the static nature of LLMs and test their adaptability to ever-changing knowledge through human evaluations. They develop a reliable evaluation protocol that uses a two-mode system: RELAXED and STRICT for a comprehensive understanding of model performance, ensuring that answers are confident, definitive, and accurate. They also provide a strong baseline named FRESHPROMPT, which seeks to enhance LLM performance by integrating real-time data from search engines.

Several benchmarks [60, 66, 165, 185] also encompass subsets that extend beyond the realm of factual knowledge or factuality. In this work, we specifically emphasize and focus on those subsets related to factuality.

Some studies [115, 125] also provide a small dataset, but they mainly concentrate on the evaluation metrics or methods for factuality, we choose to discuss them in the next subsection.

Comparison: Each benchmark has its unique strengths and limitations: **MMLU** [60]: Strength—Comprehensive coverage across diverse subjects. Limitation—Focuses on static knowledge, may not reflect real-time factual accuracy. **TruthfulQA** [97]: Strength—Specifically designed to assess truthfulness and avoid common misconceptions. Limitation—Limited to multiple-choice format, may not capture nuanced factual understanding. **HaluEval** [92]: Strength—Focuses on detecting hallucinations, a critical aspect of factuality. Limitation—Primarily based on ChatGPT, may not generalize to all LLMs. **BigBench** [165]: Strength—Extensive task diversity beyond just factual knowledge. Limitation—Complexity may make it difficult to isolate factuality-specific performance. **C-Eval** [66]: Strength—Comprehensive evaluation in Chinese context. Limitation—May not be directly applicable to non-Chinese LLMs. **SelfAware** [202]: Strength—Tests LLM’s ability to recognize knowledge boundaries. Limitation—Focused on binary (knowable/unknowable) classification, may miss gradients of certainty. **Pinocchio** [63]: Strength—Tests various aspects of factual reasoning. Limitation—High complexity may make it challenging to diagnose specific factuality issues. **REALTIMEQA** [82]: Strength—Evaluates up-to-date knowledge. Limitation—Requires frequent updates, making long-term comparisons challenging. **FreshQA** [180]: Strength—Tests adaptability to changing knowledge. Limitation—Human evaluation may introduce subjectivity.

Key findings from comparing these benchmarks: Most benchmarks focus on static knowledge, with REALTIMEQA [82] and FreshQA [180] being notable exceptions addressing the critical need for evaluating up-to-date factual knowledge. There’s a tradeoff between comprehensiveness (e.g., MMLU [60] and BigBench [165]) and specificity (e.g., TruthfulQA [97] and HaluEval [92]) in factuality evaluation. The ability of LLMs to recognize their own knowledge boundaries (as tested by SelfAware [202]) is increasingly recognized as a crucial aspect of factuality. Language-specific

benchmarks like C-Eval [66] highlight the importance of cultural context in factuality evaluation. There's a growing trend towards more dynamic and adaptive eval.

3.2 Factuality Evaluation Studies

In this section, we delve into studies that evaluate factuality in LLMs without introducing a specific benchmark, focusing primarily on those whose main contribution lies in the evaluation methodology. We spotlight works that have pioneered evaluation techniques, metrics, or have offered distinctive insights into the factuality evaluation of LLMs. We have organized the following studies based on the different types of tasks.

Question Answering: FACTOOL [19] is a tool designed to function as a factuality detector, with the primary purpose of auditing generative chatbots and assessing the reliability of their outputs. This tool is employed to evaluate several contemporary chatbots, including GPT-4, ChatGPT, Claude, Bard [55], and Vicuna [20]. Notably, FACTOOL itself leverages the capabilities of GPT-4. For the evaluation process, the researchers have curated a diverse set of prompts. The evaluation metrics included both claim-level and response-level accuracies for each chatbot. To offer a more comprehensive and equitable evaluation, a weighted claim-level accuracy is used. The weighting is determined based on the proportion of prompts from each category. The findings are illuminating. GPT-4 emerges as the top performer in terms of both weighted claim-level factual accuracy and response-level accuracy among all the chatbots assessed. Another intriguing observation is that chatbots that underwent supervised finetuning, such as Vicuna-13B, exhibited commendable performance in standard scenarios like KB-QA.

Chen et al. [16] propose CONNER, a framework that evaluates LLMs as generators of knowledge. It focuses on Factuality, Relevance, Coherence, Informativeness, Helpfulness, and Validity. It evaluates whether the generated information can be backed by external proof (Factuality), is relevant to the user's query (Relevance), and is logically consistent (Coherence). It also checks if the knowledge provided is novel or surprising (Informativeness). The Extrinsic evaluation measures whether the knowledge enhances downstream tasks (Helpfulness) and its results are factually accurate (Validity).

Wang et al. [182] task several LLMs, including ChatGPT, GPT-4 [134], BingChat [122] with answering open questions from NaturalQuestions [86] and TriviaQA [78]. They manually estimate the accuracy of those LLMs on open question answering, and find that though LLMs can achieve nice performance but still far away from perfect. Besides, they evaluate whether the GPT-3.5 can assess the correctness of LLM-generated responses, and find negative results, even if the golden answer is also presented. Similarly, Fu et al. [45] ask LLMs, such as GPT-2 and GPT-4, to directly score the factuality of a summary, and find no significant correlation between LLM's factuality indicators and human evaluations.

RAGAS [40] proposes a framework for the reference-free evaluation of **Retrieval-Augmented Generation (RAG)** systems in LLMs. This framework primarily assesses the RAG system's ability to identify relevant and key context paragraphs, the LLM's fidelity in utilizing these paragraphs, and the overall quality of the generated content. The framework focuses on Faithfulness, Answer Relevance, and Context Relevance. The authors created a WikiEval dataset, consisting of question-context-answer triplets with human judgments. This allows for the measurement of how closely different evaluation frameworks align with human assessments of faithfulness, answer relevance, and context relevance.

Generation: SelfCheckGPT [115] uses an evaluation process that encompasses several key steps. Initially, synthetic Wikipedia articles are generated using GPT-3, focusing on individuals

from the Wikibio dataset. Subsequently, manual annotation is performed at the sentence level, categorizing sentences as “Major Inaccurate”, “Minor Inaccurate”, or “Accurate”, with “Major Inaccurate” denoting sentences unrelated to the topic. Passage-level scores are derived by averaging sentence-level labels, and identifying cases of total inaccuracies through score distribution analysis. Inter-annotator agreement is assessed using Cohen’s κ [25]. Evaluation metrics primarily employ precision-recall curves (PR-Curves), distinguishing between “Non-Factual Sentences”, “Non-Factual* Sentences” (a specific subset), and “Factual Sentences.” These PR-Curves elucidate the tradeoff between precision and recall for different detection methods.

Varshney et al. [177] highlight a common real-world occurrence where users often ask questions based on false premises. These questions are challenging for state-of-the-art models. This observation necessitated the creation of a new evaluation dataset. To address this, the authors have conducted a case study and compiled a set of 50 such adversarial questions, all of which the GPT-3.5 model answered incorrectly. In order to enhance the evaluation, corresponding true premise questions were created for each of the false premise questions. This allows for a holistic evaluation of the model’s performances, taking into consideration both correct and incorrect premises. The authors make sure to evaluate the complete answers given by the model for correct and incorrect questions - in this context, it’s not enough for an answer to be partially correct, the entire answer needs to be accurate to be marked as correct.

Self-Evaluation: Kadavath et al. [79] investigate whether language models can evaluate the accuracy of their own assertions and predict which questions they can answer correctly. It is found that larger models are well-calibrated on diverse multiple-choice and true/false questions if given in the appropriate format. The approach to self-evaluation on open-ended tasks is to ask models to initially suggest answers, and then evaluate the probability ($P[\text{True}]$) that their answers are correct.

Knowledge Probing: Pezeshkpour [145] proposes a new metric for measuring whether a certain type of knowledge is present in LLM. This metric is based on information theory and measures knowledge by analyzing the probability distribution of predictions made by LLM before and after injecting target knowledge.

Model Editing: In the realm of factuality evaluation, the Model Editing task holds a unique position, focusing on refining the internal knowledge of models. This task comes with its own set of specialized evaluation metrics. Predominantly, the **Zero-Shot Relation Extraction (zsRE)** [90] and CounterFact [120] serve as the primary benchmarks for assessing Model Editing techniques. When evaluating these methods, several key criteria emerge: Reliability: Post-editing, the model should consistently generate the intended output. Generalization: The model should be adept at producing the target output even when presented with paraphrased inputs. Locality: Edits should be localized, ensuring that facts not related to the specific edit remain intact.

Multilingual Factuality Evaluation Studies: Taking into account the aspect of multilingual factual knowledge, X-FACTR [76] and MLAMA [83], which are derived from LAMA [141], evaluate the relational knowledge of models using a “fill-in-the-blank” task. Additionally, X-Fact [57] releases a benchmark for multilingual fact-checking, which involves a factual correctness classification task that encompasses a variety of topics and spans 25 typologically diverse languages from 11 different language families. Qi et al. [148] presents an extension of MLAMA and X-FACTR along with a novel metric to assess the cross-lingual consistency of language models. Shafayat et al. [157] adapts FActScore for a multilingual context by translating biographies into English. Vu et al. [179] introduce a new dataset for FActScore on texts generated by strong multilingual LLMs and find that no LLM produces consistent and reliable FActScore across languages with varying resource

levels. They also highlight the importance of the knowledge source in evaluating FActScore, showing that using the Internet as a knowledge source improves FActScore estimation in medium and low-resource languages.

Comparison. : Task Diversity: Studies cover a range of tasks, from open-domain question answering (e.g., [182]) to content generation (e.g., SelfCheckGPT [115]) and self-evaluation (e.g., Kadavath et al. [79]), reflecting the multifaceted nature of factuality in LLMs. **Evaluation Methodologies:** There's a trend towards more comprehensive evaluation frameworks, such as CONNER [16] and RAGAS [40], which assess multiple aspects of factuality beyond mere accuracy. **Human vs. Automated Evaluation:** While some studies rely on manual evaluation [182], others propose automated metrics (e.g., FACTOOL [19]). This highlights the ongoing challenge of balancing scalability with the nuanced understanding human evaluators provide. **LLM Self-Assessment:** Studies like Kadavath et al. [79] explore LLMs' ability to assess their own factuality, a crucial aspect for real-world applications. **Adversarial Testing:** Varshney et al. [177] introduce adversarial questions based on false premises, revealing LLMs' vulnerabilities to such inputs. **Context Consideration:** RAGAS emphasizes the importance of evaluating not just the answer, but also the relevance of the retrieved context, crucial for retrieval-augmented systems. **Knowledge Probing:** Pezeshkpour [145] introduces a novel metric based on information theory, offering a new perspective on measuring LLMs' knowledge.

Key findings: Even advanced LLMs like GPT-4 are not perfect in factuality, indicating significant room for improvement. LLMs struggle with self-assessment of factuality, particularly when dealing with complex or nuanced information. The performance of LLMs can vary significantly across different types of factual tasks, with some models excelling in standard QA but struggling with more complex queries. Evaluating factuality in generative tasks remains challenging, requiring sophisticated metrics and often human validation. The ability of LLMs to handle false premises or contradictory information is still limited, pointing to areas for future research and development.

4 Mechanism of Factuality

In the previous Section 3, we present quantitative statistics related to evaluating factuality. In this section, we delve deeper, exploring the underlying mechanisms that influence factuality in LLMs.

4.1 Analytical Studies of Factuality

This subsection delves into intriguing analyses concerning the factuality of LLMs, focusing on mechanisms beyond direct evaluation or enhancement. Specifically, we investigate the mechanisms through which LLMs process, interpret, and produce factual content. The following sections offer an in-depth examination of different dimensions of factuality in LLMs, ranging from their knowledge storage, and self-awareness to their approach to managing conflicting data.²

4.1.1 Knowledge Storage. The language model serves as a repository of knowledge, encoding a multitude of information about the world within its parameters [143]. However, the organization of this knowledge within LLMs remains largely mysterious. [120] introduce a methodology called causal tracing to measure the indirect impact of hidden states or activations. This technique was employed to illustrate that factual knowledge is primarily stored in the early-layer **feed-forward networks (FFNs)** of such models. Similarly, Geva et al. [53] also suggests that a substantial portion of factual information is encoded within the FFN layers. They conceptualize the input of the FFN as a query, the first layer as keys, and the second layer as values. Consequently, the intermediate hid-

²While some research [130, 183] applies some similar analyses to Pretrained Language Models. This review excludes them as they are not typically considered LLMs.

den dimension of the FFN can be interpreted as the number of memories within the layer, and the intermediate hidden state represents a vector comprising activation values for each memory. As a result, the final output of the FFN can be understood as the weighted sum of activated values. The authors further demonstrate that the value vectors often encapsulate human-interpretable concepts and knowledge [50, 51]. In addition, Chen et al. [18] have made an intriguing finding that the language model contains language-independent neurons that express multilingual knowledge and degenerate neurons that convey redundant information by applying the integrated gradients method [111]. Nevertheless, the aforementioned studies primarily focus on the representation of individual facts, and the comprehensive understanding of how factual knowledge is precisely organized and interconnected within these models remains an ongoing challenge.

4.1.2 Knowledge Completeness and Awareness. This subsubsection explores the intriguing realm of LLMs' self-awareness, their capacity to discern their knowledge gaps, and the interplay between their internally generated knowledge and externally retrieved information. We delve into the distinction between parametric knowledge and retrieved knowledge, exploring the promises and challenges these models present in knowledge-intensive tasks.

Knowledge Awareness. Several studies have investigated the knowledge awareness of LLMs, specifically assessing whether LLMs can accurately estimate the correctness of their own responses. Most of these studies treat LLMs as "black boxes," prompting the models to report their confidence levels or calculating the perplexity of the model's output as an indicator of response likelihood. Gou et al. [56] explore the model's ability to validate and iteratively refine its outputs, akin to how humans interact with tools. The authors find that solely relying on self-correction without external feedback can lead to marginal improvements or even diminished performance. [154] experiment with settings either augmented or not with external document retrieval to determine whether models recognize their own knowledge boundaries. Their findings indicate that LLMs possess an inaccurate perception of their factual knowledge boundaries and tend to be overly confident about their responses. LLMs often fail to fully harness the knowledge they possess; however, retrieval enhancement can somewhat compensate for this shortcoming. Yin et al. [202] introduce a dataset named "SelfAware" to test if models recognize what they don't know, encompassing both answerable and unanswerable questions. The experiment suggests that models do possess some capacity to discern their own knowledge gaps, but they are still far from human levels. GPT-4 outperforms other models, instructions and In-Context-Learning [36] can enhance a model's discriminatory ability. Kadavath et al. [79] focus on LLM self-assessment based on Language Model calibration using multiple-choice questions. Their findings revealed that the "none of the above" option decreased accuracy, larger models showed better calibration, and RLHF hindered model calibration levels. However, simply adjusting the temperature parameter can rectify this issue. Azaria and Mitchell [5] assess the truthfulness of statements generated by LLMs, by using the model's internal state and hidden layer activations. The authors, employing a feedforward neural network, can classify if the model is misleading by utilizing the hidden output states.

Parametric Knowledge vs. Retrieved Knowledge. [203] explore whether the internal knowledge of LLMs can replace the retrieved documents on knowledge-intensive tasks. They ask LLMs, such as InstructGPT, to directly generate contexts given a question rather than retrieving them from the database. They find the generated documents contain the golden answers more often than the top retrieved documents. Then they feed the generated docs and retrieved docs to the Fusion-in-Decoder model [71] for knowledge-intensive tasks such as Open-domain QA [86] and find the generated docs are more effective than the retrieved docs, suggesting that the LLMs contain enough knowledge for knowledge-intensive tasks. On the contrary, these observations have

been contested in subsequent investigations. Kandpal et al. [80] underscore the dependency of LLMs on the number of associated documents seen during pre-training. They argue that the success in answering fact-based questions is highly linked to the number of documents containing the topic of the question that were encountered in pre-training. The study further posited the necessity of scaling models extensively to achieve competitive performance for questions with minimum representation in the training data. Adding to these concerns, Sun et al. [167] critically evaluate the factual knowledge base of LLMs, using a specifically designed Head-to-Tail benchmark comprised of 18K question-answer pairs. The results show that the understanding of factual knowledge, particularly related to torso-to-tail entities, by currently available LLMs is suboptimal.

4.1.3 Contextual Influence and Knowledge Conflict. This sub-subsection examines the interplay between an LLM's inherent parametric knowledge and the provided contextual knowledge, exploring the capacity to utilize context and its behavior when confronted with conflicting information.

Contextual Influence on Generation. Some works explore the model's capacity to utilize context, for example, Li et al. [91] observe that larger models tend to rely on their parametric knowledge, even when faced with counterfactual contexts. This suggests that as models increase in size, they might grow more confident in their internal knowledge, potentially sidelining external context. However, the introduction of irrelevant contexts can still influence their outputs. The balance between controllability (relying on relevant context) and robustness (resisting irrelevant context) emerges as a challenge in LLM training. The study indicates that reducing context noise improves controllability, but the effect on robustness remains to be seen. In contrast, Zhou et al. [217] propose prompt templates to guide LLMs towards more faithful text generation. Among these, opinion-based prompts prove most effective, indicating that when LLMs are queried about opinions, they adhere more closely to the context. Interestingly, the study finds that using counterfactual context enhances the model's faithfulness, while original context, sourced from platforms like Wikipedia, might induce a simplicity bias, leading LLMs to answer questions without heavily relying on the context. Chen et al. [15] conduct a comprehensive evaluation of LLMs' ability to effectively utilize retrieved information. The study reveals that while retrieved documents can boost LLM performance, the presence of noise in these documents can hinder it. Yue et al. [205] investigate the nature of LLM-generated content in relation to provided references. They categorize the generated content as attributable, contradictory, or extrapolatory to the reference. Both fine-tuned models and instruction-based LLMs struggle to accurately evaluate the alignment between generated content and references, underscoring the challenge of ensuring that LLMs produce content consistent with the provided context. Shi et al. [158] study the distractibility of LLMs on the GSM-IC dataset derived from GSM8K [24]. They discover that all the prompting techniques are responsive to irrelevant information in the problem definition. They identify various factors of irrelevant information that impact the model's sensitivity to irrelevant context. Moreover, they find that self-consistency prompting and incorporating irrelevant information into the exemplars can enhance models' performance, enabling them to learn to disregard irrelevant information.

Handling Knowledge Conflicts. A series of studies are interested in LLMs' behavior when confronted with conflicting information. Longpre et al. [108] introduce the concept of knowledge conflicts, where the provided context contradicts the model's learned information. Their findings suggest that such conflicts lead to increased prediction uncertainty, especially for in-domain examples. Observations across models, ranging from T5-60 M to 11 B, indicate that larger models tend to prioritize their parametric knowledge. Moreover, there's an inverse relationship between retrieval quality and the tendency to rely on internal knowledge: the more irrelevant the evidence, the more the model defaults to its parametric knowledge. [194] explore the behavior of recent LLMs,

including ChatGPT and GPT-4. Contrary to findings on smaller LMs, they discover that LLMs can be highly receptive to external evidence, even if it contradicts their parametric memory, provided the external evidence is coherent and convincing. Additionally, LLMs exhibit a strong confirmation bias, especially when presented with evidence that aligns with their parametric memory. This bias becomes even more pronounced for widely accepted knowledge. In scenarios where no relevant evidence is provided, LLMs tend to express uncertainty. However, when presented with both relevant and irrelevant evidence, they demonstrate an ability to filter out irrelevant information. Wang et al. [189] argue that LLMs should not rely solely on either parametric or non-parametric information, but grant LLM users the agency to make informed decisions. They introduce a framework including three tasks ((1) Contextual knowledge conflict detection; (2) QA-span knowledge conflict detection; (3) Distinct answers generation) to simulate knowledge conflicts and evaluate whether LLMs' behaviors align with the goal.

4.2 Causes of Factual Errors

Understanding the root causes of these factual inaccuracies is crucial for refining these models and ensuring their reliable application in real-world scenarios. In this subsection, we delve into the multifaceted origins of these errors, categorizing them based on the stages of model operation: Model Level, Retrieval Level, Generation Level, and other miscellaneous causes. Table 1 shows examples of factuality errors caused by different factors.

4.2.1 Model-level Causes. This subsection delves into the intrinsic factors within LLMs that contribute to factual errors, originating from their inherent knowledge and capabilities.

Domain Knowledge Deficit. The model may lack comprehensive expertise in specific domains, leading to inaccuracies. Every LLM has its limitations based on the data it was trained on. If an LLM hasn't been exposed to comprehensive data in a specific domain during its training, it's likely to produce inaccurate or generalized outputs when queried about that domain. For instance, while an LLM might be adept at answering general science questions, it might falter when asked about niche scientific subfields [110].

Outdated Information. The model's dependence on older datasets can make it unaware of recent developments or changes. LLMs are trained on datasets that, at some point, become outdated. This means that any events, discoveries, or changes post-dating the last training update won't be known to the model. For example, ChatGPT and GPT-4 are both trained on data up to 2021.09 might not be aware of events or advancements after then.

Immemorization. The model does not always retain knowledge from training corpus. While it's a misconception that LLMs "memorize" data, they do form representations of knowledge based on their training. However, they might not always recall specific, less-emphasized details from their training parameters, especially if such details were rare or not reinforced through multiple examples. For example, ChatGPT has pretrained with Wikipedia, but it still fail in answering some questions constructed from Wikipedia [182].

Forgetting. Models may fail to retain knowledge from their training phase or lose prior knowledge during further training. As models undergo additional training on new data, there's a risk of "catastrophic forgetting" [17, 54, 190, 206] where they might lose certain knowledge they previously know. This is a well-documented challenge in neural network training, where networks forget previously learned information when exposed to new data, a phenomenon also observed in LLMs [113].

Reasoning Failure. While models may possess relevant knowledge, they can sometimes fail to effectively reason with this information to answer queries. Even if an LLM has the requisite knowledge to answer a question, it might struggle to connect the relevant concepts or engage in logical reasoning. For instance, ambiguity in the input [100] can potentially lead to a failure in understanding by LLMs, consequently resulting in reasoning errors. In addition, Berglund et al. [8] find the LLM traps in the reversal curse, for example, it knows that A is B’s mother but fails to answer who is B’s son. This limitation is particularly evident in complex multi-step reasoning tasks or when the model needs to infer based on a combination of facts [85, 171].

4.2.2 Retrieval-level Causes. The retrieval process plays a pivotal role in determining the accuracy of LLMs’ responses, especially in retrieval-augmented settings. Several factors at this level can lead to factual errors:

Insufficient Information. When the retrieved data fails to provide enough context or details, the LLM may struggle to generate a factual response. This can result in generic or even inaccurate outputs due to the lack of comprehensive evidence.

Misinformation Not Recognized by LLMs. LLMs are prone to accept and propagate misinformation present in the retrieved data. This issue is especially concerning when the model encounters knowledge conflicts, where the retrieved information contradicts its pre-trained knowledge, or multiple retrieved documents present conflicting information [94]. For instance, [108] observed that the more irrelevant the evidence, the more likely the model is to rely on its intrinsic knowledge. Recent studies, such as [137], have also shown that LLMs are susceptible to misinformation attacks within the retrieval process.

Distracting Information. LLMs can be misled by irrelevant or distracting information in the retrieved data. For example, if the evidence mentions a “Russian movie” and “Directors”, the LLM might incorrectly infer that “The director is Russian”. [112] highlighted this vulnerability, noting that LLMs can be significantly impacted by distracting retrieval results. They further proposed instruction tuning as a potential solution to enhance the model’s ability to sift through and leverage retrieval results more effectively.

Additionally, when dealing with long retrieval inputs, the models tend to show their best performance when processing information given at the beginning or end of the input context, according to Liu et al. [103]. In contrast, the models are likely to experience a significant decrease in performance when they are required to extract relevant data from the middle of these extensive contexts.

Misinterpretation of Related Information. Even when the retrieved information is closely related to the query, LLMs may occasionally misunderstand or misinterpret it. Although this issue becomes less prevalent with optimized retrieval processes, it remains a potential source of error. For instance, the rate of errors dropped significantly when the retrieval process was improved [199].

4.2.3 Inference-level Causes.

Snowballing. During the generation process, a minor error or deviation at the initial stages can propagate and amplify as the model continues to generate subsequent content. For instance, if an LLM misinterprets a prompt or starts with an inaccurate premise, the subsequent content can be further from the truth [177, 208].

Erroneous Decoding. The decoding phase is crucial for translating the model’s internal representations into human-readable content [22, 116]. Errors During the phase, whether stemming from issues like beam search limitations or suboptimal sampling strategies, can result in outputs

that misrepresent the model's actual knowledge or intention. Furthermore, modifying the decoding process for safety or legal compliance [68] has been shown to reduce the factual accuracy of LLMs [105]. This can manifest as inaccuracies, contradictions, or even nonsensical statements.

Exposure Bias. LLMs are inherently shaped by their training data. When certain types of content or phrasing appear more frequently in the training corpus, LLMs may exhibit a propensity to generate similar content, even if it is not the most accurate or pertinent. This bias can be especially pronounced if the training data has imbalances or if certain factual scenarios are underrepresented [43, 47]. Consequently, the model's outputs, in such cases, reflect its training exposure rather than objective factuality. For example, research conducted by [61] suggests that LLMs can correctly identify the gender of individuals who conform to the binary gender system, but they perform poorly when tasked with determining non-binary or gender-neutral identities.

4.3 Summary

4.3.1 Meaningful Achievements. **Knowledge Storage:** Research has shown that factual knowledge is primarily stored in the FFNs of early layers in LLMs [53, 120]. This finding provides crucial insights into the internal structure of LLMs. **Multilingual Knowledge:** The discovery of language-independent neurons that express multilingual knowledge [18] has significant implications for developing multilingual LLMs. **Self-Awareness:** Studies have demonstrated that LLMs possess a degree of self-awareness, capable of recognizing their own knowledge boundaries [202]. **Context Utilization:** Methods to enhance LLMs in utilizing context have been discovered, such as using counterfactual contexts and opinion-based prompts [217]. **Knowledge Conflict Handling:** Larger LLMs have been found to perform better in handling conflicting information, balancing parametric knowledge with external evidence [194].

4.3.2 Future Directions. **Comprehensive Knowledge Organization:** While we understand the representation of individual facts, a comprehensive understanding of how LLMs organize and interconnect complex knowledge networks is still lacking. **Dynamic Knowledge Updating:** More research is needed to explore how to effectively update LLMs' knowledge to avoid outdated information and forgetting issues. **Reasoning Mechanisms:** Although LLMs perform well on some reasoning tasks, we still don't fully understand their reasoning mechanisms, especially in complex multi-step reasoning tasks. **Bias Mitigation:** Further research is needed on how to reduce exposure bias and other types of biases to improve the fairness and accuracy of LLMs. **Retrieval Optimization:** While retrieval-augmented methods show promise, more research is needed to optimize the retrieval process, especially in handling long text inputs and filtering irrelevant information. **Cross-modal Factuality:** With the development of multimodal LLMs, research is needed on maintaining factual consistency across text, images, and other modalities.

These achievements and research gaps highlight the complexity and importance of factuality research in LLMs, while also providing guidance for future research directions.

5 Enhancement of Factuality

This section discusses methods to enhance factuality in LLMs across different phases, including LLM generation, RAG, inference-phase enhancements, as illustrated in Figure 2. It's essential to recognize that various research articles may employ distinct experimental settings, such as zero-shot, few-shot, or full settings. Therefore, when interpreting this table, readers should be aware that performance metrics for different methods, even when evaluating the same metric on the same dataset, may not be directly comparable.

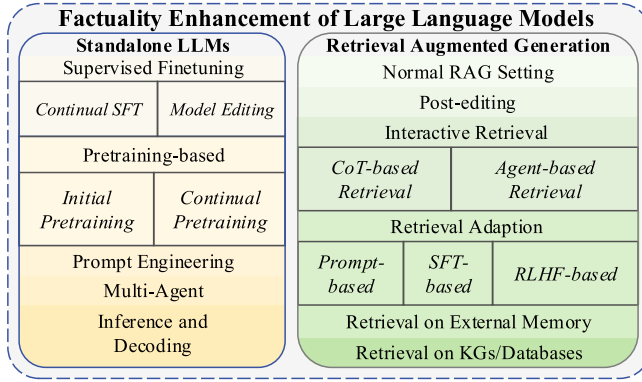


Fig. 2. An overview of methods to enhance factuality in LLMs. This encompasses three primary areas: enhancement techniques for pure LLMs and strategies for retrieval-augmented LLMs.

5.1 On Standalone LLM Generation

When focusing on standalone LLM generation, enhancement strategies can be broadly grouped into three main categories: (1) *Improving Factual Knowledge from Unsupervised Corpora* (Section 5.1.1): This involves refining the training data during pretraining, such as through deduplication and emphasizing informative words [87]. Techniques like TOPICPREFIX [88] and sentence completion loss are also explored to enhance this approach. (2) *Enhancing Factual Knowledge from Supervised Data* (Section 5.1.2): Examples in this category include supervised finetuning strategies [23, 214] focus on finetuning with labelled data or integrating structured knowledge such as **knowledge graphs (KGs)** or making precise adjustments to model parameters [93]. (3) *Optimally Eliciting Factual Knowledge from the Model* (Sections 5.1.3–5.1.5): This category encompasses methods like Multi-agent collaboration [37] and innovative prompts [203]. Additionally, novel decoding methodologies, such as factual-nucleus sampling, are introduced to further improve factuality [22, 88]. Some work [101, 169] focus on improving the factuality of large multi-modal models, we choose to not emphasize them.

5.1.1 Pretraining-based. Pretraining plays a pivotal role in equipping the model with the factual knowledge derived from the corpus. By emphasizing strategies during this phase, the model's inherent factuality can be significantly enhanced. This approach is particularly crucial for addressing challenges like immemorization and forgetting.

Initial Pretraining. Methods employed during the foundational pretraining of the model. Lee et al. [87] develop two distinct tools for deduplicating training data, addressing the issue of redundant texts and long repeated substrings present in the training sets of current LLMs. These tools effectively mitigate the recall of memorized texts in model outputs. Remarkably, they achieve similar or even superior accuracy with fewer training steps. Sadeq et al. [155] introduce a modification to the Masked Language Model training objective used in the pretraining of LLMs. They discover that high-frequency words do not consistently contribute to the model's acquisition of factual knowledge. To address this, they devise a strategy that encourages the language model to prioritize informative words during unsupervised training. This is achieved by masking tokens more frequently based on their informative relevance. To quantify this relevance, they utilize **Pointwise Mutual Information (PMI)** [42], positing that words with higher PMI values, in relation to their adjacent words, are likely to be more informatively significant.

Continual Pretraining. Iterative pretraining processes that allow the model to progressively refine and update its knowledge base. Lee et al. [88] introduce TOPICPREFIX as a pre-processing method and the sentence completion loss as training objective. Some factual sentences may be unclear when the LM training corpus is chunked, especially when these sentences contain pronouns (e.g., she, he, it). So they prepend TOPICPREFIX (e.g., Wikipedia document name) to sentences in the factual documents to transform each sentence into an independent factual statement. They also introduce the sentence completion loss, with the aim of enabling the model to capture facts from entire sentences rather than just focusing on the associations between subwords. For implementation, they establish a pivot t for each sentence and require zero-masking for all token prediction losses before t . This pivot is only necessary during the training phase.

5.1.2 Supervised Finetuning. Supervised finetuning leverages labeled datasets to enhance the model's performance. This approach serves a dual purpose: it imparts specific task or knowledge base-oriented information to the model and addresses challenges like memorization and forgetting. Several studies, such as Chung et al. [23], Zhou et al. [214], emphasize the pivotal role of supervised finetuning in eliciting the inherent knowledge of the base model, subsequently enhancing its reasoning capabilities.

Continual SFT. A cyclic finetuning approach, where the model undergoes consistent refinement using sequential sets of labeled data.

Moiseev et al. [128] investigate how to inject structured knowledge from a KG into LLMs. The approach involves directly training T5 using triplets containing relationship knowledge. Previous methods often describe the triplets using prompts and then train LLMs with masked language model task, but some triplets are not easy to describe. They compare three knowledge-enhancement finetuning methods, which are MLM training on the C4 corpus [152], masking the subject or object in KG triplets, and masking the subject or object in the KELM corpus [1].

FactTune [135] aims to use Reinforcement Learning to improve the factuality, necessitating scoring the factuality of the content generated by the model is needed. Given the high cost of human labeling, authors implement two alternative methods to assess the factuality of LLM-generated content: (1) Reference-based approach, utilizing external knowledge base retrieval and Factscore [125] method to score; (2) Reference-free approach, leveraging another LLM's confidence to score. Then, they utilize the Directly Preference Optimization [150] algorithm to optimize the model.

Sun et al. [168] introduce negative samples and contrastive learning to the supervised finetuning process with MLE loss. The sources of these negative samples are either from the KG or are generated by LLMs. While traditional contrastive learning can only function at the token or sentence level, the intrinsic value of span information cannot be overlooked, they employ Named Entity Recognition to extract this critical span data. The training process integrates a combination of MLE loss, standard contrastive learning loss, and span-based contrastive learning loss, with carefully calibrated parameters to maximize performance. Yang et al. [198] presents a development framework for **Knowledge Graph-enhanced Large Language Models (KGLLMs)**, drawing from established technologies. They elaborate on various enhancement strategies, including a before-training enhancement that refines input quality by integrating factual data; a during-training enhancement that synergizes textual and structural knowledge, utilizing tools like graph neural networks and attention mechanisms; multi-task learning which focuses on knowledge-guided pre-training tasks to bolster the factual knowledge acquisition of LLMs; and a post-training enhancement that fine-tunes LLMs for domain-specific tasks using knowledge-rich data. Additionally, the significance of prompt learning in LLMs is highlighted, emphasizing the importance of choosing appropriate prompt templates.

Model Editing. Instead of directly finetuning the model, model editing is a more precise approach to enhance the model’s factuality. By editing specific areas that are related to the fact, the model can correctly express that fact without compromising other unrelated knowledge. Current editing methods [200] can be categorized into weight-preserved and weight-modified paradigms. When modifying the model’s weight, KN [29] and ROME [120] initially analyze representations to locate those underlying factual errors and then directly update the relevant weights. Meanwhile, KE [31] and MEND [126] employ a hypernetwork to learn the necessary weight adjustments. While effective, the robustness and generalization of directly updating weights remains an open question.

Li et al. [93] present a technique known as **Inference-Time Intervention (ITI)** designed to boost the factual accuracy of LLMs. ITI adjusts model activations during inference to enhance the truthfulness of responses. This method, categorized under activation editing, is both adjustable and less intrusive, distinguishing it from weight editing techniques. Drawing inspiration from prior research, they utilize steering vectors, which are proven effective for style transfer, to guide model activations. They detail a multi-step model selection process involving the calibration of intervention strength hyperparameters, pinpointing truth-telling related heads, and determining their truth-telling directions.

5.1.3 Multi-Agent. Engaging multiple models in a collaborative or competitive manner, enhancing factuality through their collective capabilities, helps address issues related to memorization limitations and reasoning failure problems.

Du et al. [37] propose an approach to enhance the performance of language models by treating different LLMs as intelligent agents engaged in multi-agent debates. In this method, multiple instances of language models present and debate their respective answers and reasoning processes, ultimately reaching a consensus on the final answer after multiple rounds of debate. If the debate’s answers fail to converge, prompts are modified to reduce the stubbornness of the two agents. This approach has been demonstrated to significantly improve mathematical and reasoning abilities across various tasks while enhancing the factual accuracy of generated content.

Cohen et al. [26] develop a fact-checking mechanism. Drawing parallels to a scenario where a witness is interrogated for the veracity of their claims, they utilize a LLM to gather statements from a QA dataset, which are either factually correct or incorrect. During the statement generation, the model is provided with a golden answer, prompting it to produce both accurate and inaccurate statements, inherently labeling each statement. For every QA pair, another LLM, acting as an interrogator, generates a series of questions. A separate LLM, playing the role of the respondent, answers these queries. This iterative questioning and answering continues until the interrogator is satisfied, culminating in a conclusion.

5.1.4 Novel Prompt. Introducing innovative or tailored prompts to extract more factual and precise responses from the LLM, can better assist the model, elicit the knowledge in its parameters and improve the reasoning ability. Yu et al. [203] introduce a novel approach called **Generate-then-Read (GENREAD)**. Document retrievers are replaced by LLM generators. In this article, the LLM is prompted to generate multiple contextual documents on a given question. The authors then cluster these document embeddings, and sample documents from different clusters to ensure diversity of contextual documents. With these generated in-context demonstrations, LLMs achieve better results on knowledge-intensive tasks than retrieving from external corpus such as Wikipedia. Weller et al. [192] introduce a metric called QUIP-Score to measure the dependency on pre-trained data. They index Wikipedia to swiftly determine the extent to which an LLM’s response relies on its pre-training. By using specific prompts, like “Based on evidence from Wikipedia:” they aim to evoke the LLM’s recall of content from its training dataset. In addition to grounding prompts, they also introduce anti-grounding prompts to encourage the

LLM to respond without referencing its training data, for instance, “Respond without using any information from Wikipedia.” The motivation behind this approach is the belief that guiding the LLM to reference more of the knowledge it acquires during pre-training can reduce the generation of incorrect information. To quantify this grounding, they propose the QUIP-Score metric to gauge the similarity between the model’s generated content and the most relevant content in Wikipedia.

Khot et al. [84] present Decomposed Prompting. Complex tasks are broken down into multiple simpler tasks via prompting and ,which can be addressed by task-specific LLMs. For instance, for tasks involving extremely long input sequences, this technique systematically decomposes the input into shorter sequences for individual processing.

Dhuliawala et al. [34] present **Chain-of-Verification (CoVe)** to reduce factual errors. The CoVe strategy involves the model initially crafting a response, subsequently formulating verification queries to assess its initial draft, independently responding to these queries to maintain unbiased answers, and ultimately producing a validated reply. The CoVe method encompasses four pivotal stages: (1) Drafting an initial reply based on a query using the LLM, (2) Generating verification questions from the query and initial answer to pinpoint potential errors, (3) Responding to each verification question and comparing these answers with the initial response to detect discrepancies, and (4) If discrepancies are found, producing a revised answer that integrates the verification outcomes. The entire procedure is executed by prompting the same LLM differently to achieve the intended results.

5.1.5 Decoding. Decoding methodologies, such as beam search and nucleus sampling, play a crucial role in directing the model to produce outputs that are both factual and coherent. By refining the decoding process, challenges like snowballing errors or erroneous decoding, as detailed in Section 4.2.3, can be effectively addressed.

Lee et al. [88] propose a new decoding sampling algorithm called factual-nucleus sampling that achieves a better tradeoff between generation quality and factuality when compared to prevailing decoding algorithms. They postulate that the randomness of sampling is more harmful to factuality when applied to the generation of the latter portion of a sentence as opposed to its initial segment. So the factual-nucleus sampling algorithm, an adaptation of nucleus sampling, can dynamically adjust the ‘nucleus’ probability throughout the generation of each sentence, progressively reducing randomness with each successive generation step. The ω -bound parameter is provided to prevent the p-value from becoming too small and hurting diversity.

Chuang et al. [22] propose “Decoding by Contrasting Layers” to mitigate these hallucinations. This approach leverages the differences in logits obtained from projecting the later layers versus earlier layers to the vocabulary space, taking advantage of the known localization of factual knowledge in LLMs. The results of this study demonstrate that DoLa consistently enhances the truthfulness of LLM-generated content across various tasks, such as multiple-choice and open-ended generation tasks, showcasing its potential to significantly improve the reliability of LLMs in generating accurate and truthful facts.

5.2 On RAG

RAG has emerged as a widely adopted approach to address certain limitations inherent to standalone LLMs, such as outdated information and the inability to memorize [13, 102]. These challenges are elaborated upon in Section 4.2.1. Yet, while RAG offers solutions to some issues, it introduces its own set of challenges, including the potential for insufficient information and the misinterpretation of related data, as detailed in Section 4.2.2. This subsection delves into various strategies devised to mitigate these challenges. Within the realm of RAG, enhancement techniques

can be broadly categorized into several pivotal areas: (1) *The Normal Setting of Utilizing Retrieved Text for Generations* (Section 5.2.1): (2) *Interactive Retrieval and Generation* (Section 5.2.2): Examples here include the integration of Chain-of-Thoughts steps into query retrieval [59] and the use of an LLM-based agent framework that taps into external knowledge APIs [199]. (3) *Adapting LLMs to the RAG Setting* (Section 5.2.3): This involves methods like the one proposed by Peng et al. [140], which combines a fixed LLM with a plug-and-play retrieval module. Another notable approach is REPLUG [159], a retrieval-augmented framework that considers the LLM as a black box and finetunes retrieval models using language modeling scores. (4) *Retrieving from Additional Knowledge Bases* (Sections 5.2.5 and 5.2.4): This category includes methods that retrieve from external parametric memories [14] or KGs [209] to enhance the model's knowledge base.

5.2.1 Normal RAG Setting.

Workflow of Normal RAG Setting. A normal RAG setting works by retrieving external data and incorporating it into a LLM during the generation phase. We follow the framework proposed by LlamaIndex [102] and LangChain [13] to decouple the process into the following modules and steps: (1) *Document loaders* are used to import documents from various sources. These loaders can fetch different types of documents (HTML, PDF, code) from various locations (private s3 buckets, public websites). (2) *Document transformers* are employed to extract relevant parts of the documents. This may involve segmenting or chunking large documents into smaller chunks. Different algorithms for this task are employed, optimized for specific document types like code or markdown. (3) *Text embedding models* are employed to capture the semantic meaning of the text. (4) *Vector stores* are implemented to efficiently store and search the embeddings. (5) *Retrievers*, such as the Parent Document Retriever, Self Query Retriever, and Ensemble Retriever, are used to retrieve the data from the database. Here, the Parent Document Retriever allows for the creation of multiple embeddings per parent document to retrieve smaller chunks while maintaining a larger context. The Self Query Retriever separates the semantic part of a query from other metadata filters, allowing for more accurate retrieval. The Ensemble Retriever enables the retrieval of documents from multiple sources or using different algorithms.

5.2.2 Interactive Retrieval. While retrieval systems are designed to source relevant information, they may occasionally fail to retrieve accurate or comprehensive data. Additionally, LLMs might struggle to recognize, or even be misled by, the retrieved content, as detailed in Section 4.2.2. Implementing an interactive retrieval mechanism can address these challenges, aiding in sourcing more appropriate information and guiding the LLM towards improved content generation. In this sub-subsection, we explore methods that employ the Chain-of-Thoughts and Agent-based approaches to achieve effective interactive retrieval.

CoT-based Retrieval. In recent studies, there is a growing interest in integrating Chain-of-Thoughts [191] steps into query retrieval. He et al. [59] introduce a method that generates multiple reasoning paths and their corresponding predictions for each query. This process involves retrieving relevant knowledge from external sources like Wikidata [178], WordNet [123], and ConceptNet [164]. The faithfulness of each reasoning path determines based on entailment scores, contradiction scores, and MPNet similarities [163] with the retrieved knowledge. The prediction with the highest faithfulness score is chosen as the final result. This method demonstrates superior performance in tasks such as commonsense reasoning [52], temporal reasoning [74], and tabular reasoning [58], outperforming the baseline CoT reasoning and self-consistency methods [188]. On a related note, Trivedi et al. [176] propose IRCoT, an innovative retrieval technique that interweaves the CoT process. In this approach, each generated sentence during the CoT combines with the ques-

tion to form a retrieval query. The subsequent reasoning step is then produced by the Language Model using both the retrieval results and the prior reasoning. This interleaving method has been found to enhance the performance of both retrieval and CoT in Open-domain QA.

Agent-based Retrieval. Using an LLM-based agent framework that leverages external knowledge APIs (including Retrieval) as tools or requesting such APIs as actions.

FLARE [77] is a dynamic solution to address the limitations of previous RAG works which either retrieve only once at the onset of generation or do so based on fixed intervals. Single retrievals are insufficient for long-form generation due to the evolving information needs during the process, and fixed intervals for previous token queries can be inappropriate, FLARE determines “when and what to retrieve” dynamically. The decision of “when” is based on whether the current sentence contains a token with a generation probability below a set threshold. If no such token is found, the sentence is accepted and moves to the next generation step; otherwise, RAG occurs. To determine “what” to retrieve, the current sentence is used as a query. To address the challenge of low-probability tokens affecting retrieval accuracy, two solutions are proposed: masking these tokens, and using LLM for generation of these tokens as queries.

Yao et al. [199] present a new framework named ReAct that integrates Chain-of-Thoughts reasoning with action. Through in-context learning, the LLM’s CoT output is transformed into descriptions of reasoning processes and action steps. Subsequently, These action descriptions are standardized and executed, with the results being incorporated into subsequent prompts.

Shinn et al. [160] propose a prompt engineering framework named Reflexion to enable LLMs to reflect on and correct previous errors. They use linguistic feedback to strengthen an agent’s actions instead of adjusting model weights. Specifically, the Reflexion Agent, an LLM, first interacts with its environment by generating an “Action” via ReAct [199] or Chain-of-Thoughts [191] in few-shot scenarios, which results in an “Observation”. This “Observation”, whether a reward, error message, or natural language feedback, provides insights on the Agent’s current “Action”. When the Agent receives a failure signal, it triggers a self-reflexion mechanism, utilizing the LLM, to summarize the reasons for the failure into its Memory module, creating a “long-term memory”. On subsequent generations, the Agent can review all past reflexion memories to prevent mistakes.

Varshney et al. [177] developed a framework to reduce factual inaccuracies in LLMs. This framework uses models to identify key entities and generate questions. During content generation, it identifies important concepts like names and locations using entity extraction or LLM directives. The confidence in these concepts is assessed using logit output values. If confidence is low, the system retrieves a document to verify the information, using a binary query approach. When validation fails, the model rectifies the error using information from the retrieved document. This method has proven effective, particularly in article generation.

Self-RAG [4], an enhancement of the RAG framework, introduces a self-reflective approach to improve the factuality of LLM outputs. It involves on-demand retrieval, where the LLM assesses the relevance and supportiveness of retrieved documents. The model generates special ‘reflection tokens’ during output, a capability developed through end-to-end training. In practice, Self-RAG decodes a retrieval token for each input and generated content segment. A ‘no’ token results in the continuation of standard output generation, whereas a ‘yes’ token initiates the document retrieval process. The model then evaluates each retrieved document’s relevance and supportiveness. Using reflection tokens, it selects the most appropriate document to integrate into the ongoing content generation, repeating this process until completion.

5.2.3 Retrieval Adaptation. Recent research [154, 182] has highlighted that merely using the retrieved information in LLMs doesn’t consistently improve their performance in answering factual questions. This underscores the importance of enabling LLMs to better adapt to the retrieved data,

thereby generating more accurate responses. In this section, we delve into various strategies that facilitate this adaptation. Specifically, we explore three methodological approaches: prompt-based methods, SFT-based methods, and RLHF-based methods.

Prompt-based. Leveraging prompts to navigate the retrieval process, ensuring the extraction of pertinent and factual data. LLM-Augmenter [140] is a system using a fixed LLM combined with a plug-and-play retrieval module to help the LLM perform better in tasks that are particularly sensitive to factual errors. This system enhances its performance by enabling the LLM to use a series of modules (e.g., allowing the LLM to interact with external knowledge) to assist the LLM in generating evidence-based results. And they use automated feedback generated by utility functions (e.g., the factuality score of a LLM-generated response) to refine LLM's candidate responses.

SFT-based. Optimizing the LLM or retrieval system through training to enhance the alignment between generation tasks and the retrieved content.

Izacard et al. [72] introduce a comprehensive architecture named ATLAS which is composed of the Contriever [69] retriever of the dual-encoder architecture and the T5 [151] language model with Fusion-in-Decoder [71]. The training objectives for the retriever consist of four components: Attention Distillation [70], where the retriever is trained on the average attention scores from the language model for each article; **End-to-end training of Multi-Document Reader and Retriever (EMDR2)** [161], which involves using the query and the top-K retrieved articles from the current retriever as input and loss computation against the standard answers to train the retriever; **Perplexity Distillation (PDist)**, where the retriever is trained to predict how much the perplexity of standard answers would improve for each document; **Leave-one-out Perplexity Distillation (LOOP)**, which trains the retriever to predict how much worse the prediction of the language model gets when removing a document from the top-K results. The LM's training objectives consist of three parts: prefix language modeling, masked language modeling, and title to section generation.

Shi et al. [159] introduce REPLUG, a retrieval-augmented framework that considers the LLM as a black box, freezes its parameters and tunes retrieval models with supervision signals using language modeling scores. In this framework, the input context and the document are encoded through the dual encoder architecture, cosine similarity is then calculated to retrieve related documents. The likelihood for each retrieved document and the language model scores are computed. Then they can update the retrieval model parameters by minimizing the KL divergence between retrieved document likelihood and the language model's score distribution.

Luo et al. [112] focus on utilizing instruction-tuning to denoise the retrieval results. They gather retrieval outcomes from various search APIs and domains, leading to the creation of a new search-grounded dataset. This dataset encompasses instructions, grounding information, and responses. Notably, it includes both pertinent results and those that are unrelated or disputed. The model needs to learn to ground on useful search results.

RLHF-based. Menick et al. [121] use **reinforcement learning from human preferences (RLHP)** to train a 280 billion parameter model named GopherCite that generates answers along with high quality supporting evidence. They firstly collect data from existing models and have it rated by humans. The data is used for finetuning and reward model training. A supervised finetuning model is trained to produce accurate quotes with proper syntax. A reward model is created to rank model outputs based on overall quality. Finally, a reinforcement learning policy is optimized to align model behavior with human preferences, improving quoting performance. The model may decline to answer when the reward model score is too low.

5.2.4 Retrieval on External Memory. Currently, most LLMs enhance their factuality by retrieving knowledge in the form of text snippets from external storage and incorporating them into the context. Some researchers are exploring the storage of knowledge in non-textual forms and integrating this knowledge into models through specialized methods. Li et al. [95] store knowledge in the form of key-value pairs in memory, G-MAP [181] does not explicitly store knowledge in storage but uses a general domain PLM as external memory, KALA [81], EaE [46], and Mention Memory [32] store encoded entities in external memory. Houlsby et al. [62] propose to add Adapter modules to finetune pre-trained deep learning models on a new task with minimal changes to the original model by inserting small, trainable modules between existing layers. Chen et al. [14] finetunes a T5 model to serve as an editor, but it introduces negative samples during finetuning.

5.2.5 Retrieval on Structured Knowledge Source. We discuss studies that retrieves on structured repositories, such as KGs and database, to source factual data during generation in this subsection. Zhang et al. [209] utilize KGs for retrieval to tackle factual errors. They observe that there can be inconsistencies between a user's request and the content in the KG. StructGPT [75] is a general prompt framework to support LLMs reasoning on structured data (e.g., KG, Table, and Database). In general, the core of this framework is that they construct the specialized interfaces to collect relevant evidence from structured data (i.e., reading), and let LLMs concentrate on the reasoning task based on the collected information (i.e., reasoning). Specially, they propose an invoking-linearization-generation procedure to support LLMs in reasoning on the structured data with the help of the interfaces. Baek et al. [6] propose to inject the factual knowledge from KGs into (large) language models (up to GPT-3.5), by retrieving the relevant facts from KGs based on their textual similarities with the input question and then injecting them as the prompt of language models.

5.3 Multilingual Factuality Enhancement

In the field of multilingual factuality enhancement, researchers have proposed various approaches to improve the factual accuracy of texts generated by multilingual LLMs. Vu et al. [179] significantly improve the accuracy of FActScore by expanding knowledge sources, utilizing the internet as a knowledge resource, and leveraging knowledge generated by LLMs. Specifically, they enhance FActScore by increasing the number of retrieved paragraphs, querying relevant information through the Google Search API, and using GPT-4 to generate answers to related questions, which are then combined with the retrieved paragraphs and annotated with GemP to improve factual accuracy across all languages. Singhal et al. [162] explore techniques such as zero-shot prompting, English chain-of-thought reasoning, cross-lingual prompting, and self-consistency decoding, where multiple answers are generated and the most consistent one is selected to improve factual verification accuracy. Furthermore, Shafayat et al. [157] using FActScore introduces a multilingual FActScore pipeline, applying FActScore across multiple languages to assess the factual accuracy of multilingual LLMs in different languages and geographical regions, while also exploring the use of non-English Wikipedia articles to enhance factual evaluation. These methods provide different perspectives and solutions for multilingual factuality enhancement, ranging from expanding knowledge sources to improving prompting techniques, all contributing to the improvement of factual accuracy in texts generated by multilingual LLMs. Furthermore, hallucinations in machine translation will be further mitigated as these methods help to ground generated translations in more reliable and contextually accurate information.

5.4 Summary

5.4.1 Meaningful Achievements. Pretraining Strategies: Techniques like TOPICPREFIX [88] and sentence completion loss [88] have shown promise in enhancing factual knowledge during

the pretraining phase. **Supervised Finetuning:** Methods for injecting structured knowledge from KGs into LLMs [128] have demonstrated effectiveness in improving factual accuracy. **Model Editing:** Techniques like ITI [93] have shown the ability to boost factual accuracy without extensive retraining. **Multi-Agent Approaches:** Systems utilizing multiple LLMs in debate-style interactions [37] have shown improvements in factual accuracy and reasoning abilities. **RAG Frameworks:** Frameworks like ATLAS [72] and REPLUG [159] have significantly enhanced LLMs' ability to leverage external knowledge. **Interactive Retrieval:** Methods like ReAct [199] and Self-RAG [4] have demonstrated the effectiveness of integrating reasoning with retrieval for improved factuality. **Decoding Strategies:** Techniques like factual-nucleus sampling [88] and Decoding by Contrasting Layers [22] have shown promise in enhancing the truthfulness of generated content.

5.4.2 Future Directions. **Long-term Knowledge Retention:** More research is needed on how to effectively update and maintain factual knowledge in LLMs over time without catastrophic forgetting. **Scalability of Model Editing:** While promising, current model editing techniques often focus on specific facts. Scaling these approaches to handle large-scale knowledge updates remains a challenge. **Evaluation of Multi-Agent Systems:** While multi-agent approaches show promise, more comprehensive evaluation frameworks are needed to assess their factual consistency across diverse domains. **Efficiency of RAG Systems:** Current RAG methods often involve significant computational overhead. Research into more efficient retrieval and integration mechanisms is needed. **Handling Conflicting Information:** More sophisticated methods are needed for LLMs to handle and reason about conflicting information from different sources. **Domain Adaptation:** While some domain-specific enhancements have been explored, more research is needed on efficiently adapting general LLMs to specialized domains without compromising their broad capabilities. **Ethical Considerations:** As factuality enhancement techniques become more sophisticated, research is needed on their potential misuse and how to ensure they are applied ethically. **Interpretability:** As enhancement methods become more complex, ensuring the interpretability of how LLMs arrive at factual conclusions becomes increasingly important. **Cross-lingual Factuality:** More research is needed on maintaining and enhancing factuality across different languages and cultural contexts.

6 Future Directions

Despite the advancements detailed in this survey, several challenges loom large. The evaluation of factuality remains an intricate puzzle, complicated by the inherent variability and nuances of natural languages. The core processes governing how LLMs store, update, and produce facts have not yet been fully revealed. And while certain techniques, like continual training and retrieval, show promise, they are not without limitations.

Evaluation. Assessing the factual accuracy of LLM outputs is a complex task that extends beyond traditional linguistic evaluation metrics [88, 145]. Current methods often fall short in determining the truthfulness of generated content, making it often show a low correlation with human preferences [21, 38]. Additionally, the probability of the first token in MCQ may not accurately reflect the overall result [187], leading to biased or incomplete assessments. The vastness of human knowledge and the subtleties of language add layers of difficulty to this evaluation process [106]. Cultural context [11], idiomatic expressions [215], and evolving terminologies further complicate automated factuality evaluation. Leveraging KGs [106, 167], truth discovery [94], and cross-referencing could enhance the evaluation process.

Interpreting Knowledge Storing Mechanisms. Understanding how LLMs store and retrieve knowledge remains a significant challenge [3]. Despite their ability to encode vast amounts of information from massive datasets, the exact mechanisms underlying their internal representations are opaque. The highly distributed nature of knowledge within the model’s parameters complicates efforts to identify where specific facts are stored or how they are accessed during inference [53, 120, 143]. Future research needs to develop new interpretability methods to unravel these complex processes, which could lead to more controllable and transparent models.

Discrepancy Between Internal Knowledge Storage and Actual Output. LLMs often exhibit a gap between the knowledge they have internalized and the information they provide in their outputs [197, 207]. This discrepancy manifests in two significant ways. First, models may fail to represent or utilize their stored knowledge accurately, resulting in outputs that are factually incomplete, ambiguous, or incorrect—even when the correct information exists within their training data [196]. Factors such as context, prompt phrasing, and inherent biases can distort the retrieval and expression of this knowledge [30]. Second, LLMs may generate confident responses even when they lack the necessary knowledge [154], leading to the creation of incorrect or misleading information—a phenomenon. Addressing these challenges requires ensuring that latent knowledge is faithfully and consistently reflected in the generated text [197, 201, 202, 207], and that models appropriately handle situations where they do not possess sufficient information.

Enhancing Factuality in LLMs. Existing approaches face significant hurdles, such as reliance on external data sources that may be incomplete or outdated [129, 149], and the potential for new information to interfere with established knowledge within the model. Efforts to reduce model-generated unfounded or incorrect information are ongoing but have yet to achieve full effectiveness. Future research may focus on integrating robust fact-verification processes within models [201], enhancing their ability to self-assess and flag uncertainties [197], and developing architectures that balance the incorporation of new knowledge with the preservation of existing information.

Fairness, Safety, and Factuality. While prioritizing trustworthiness, such as fairness and safety in LLMs is essential, these objectives can sometimes unintentionally compromise factuality. The longstanding tension between fairness and accuracy in machine learning has been well-documented [39, 104]. For example, removing biased language may also remove the context necessary for conveying the full truth. Similarly, safety measures can result in overly cautious models [105, 216], which might reject normal queries or alter/omit factual information. Furthermore, studies that modify the decoding process for safety or legal compliance [68] have also been shown to reduce the factual accuracy of LLMs [105]. This creates a challenge: balancing the ethical imperatives of fairness and safety with the need for accurate and complete information. Future research should explore methods to refine these tradeoffs, aligning LLM outputs with human values without compromising information integrity.

7 Conclusion

Throughout this survey, we have systematically explored the intricate landscape of factuality issues within LLMs. We began by defining the concept of factuality (Section 2.2). Our journey took us through the multifaceted realm of factuality evaluation, encompassing benchmarks and metrics (Section 3.1) and specific evaluation studies (Section 3.2). We then delved deeper, probing the intrinsic mechanisms that underpin factuality in LLMs (Section 4). Our exploration culminated in the discussion of enhancement techniques, both for standalone LLMs (Section 5.1) and retrieval-augmented LLMs (Section 5.2). We also highlight the limitations and future directions (Section 6).

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